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The development of maritime autonomous surface ships (MASS) from seafarers' perspective: operational, spatial, and labour implications

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Abstract

The development of maritime autonomous surface ships (MASS) is increasingly considered as a solution to labour shortages, cost pressures, and safety concerns in maritime transport. While much of the discussion centres on technical feasibility and regulatory challenges, this paper adopts a labour- and space-sensitive perspective by examining the views of current and former maritime professionals. Combining a thematic literature review with interviews of officers, a maritime pilot, and a port executive, we assess practical, spatial, and organizational implications of deploying MASS. The results highlight tensions between levels of automation and remaining manual operations, the limits of MASS adoption in complex coastal contexts, and divergent opinions regarding the future of maritime labour. We argue that the introduction of autonomous systems will not simply replace human labour but will redefine it in ways that require tailored regulatory, training, and infrastructural adaptations. Our findings, although exploratory, call for a human-centric transition framework attentive to spatial variability and occupational expertise.

Keywords Maritime autonomous surface ships · Unmanned vessels · Labour · Automation · Port operations · Officer careers

Abbreviations

MASS	Maritime autonomous surface ships
IMO	International Maritime Organization
ROC	Remote Operation Centre
SCC	Shore Control Centre
ISM Code	International Safety Management Code
FBW	Fly-by-wire
OPS	Onshore power supply/shore power supply
ENSM	École Nationale Supérieure Maritime
PSV	Platform supply vessel
RoRo	Roll-on/roll-off
AI	Artificial intelligence

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EU European Union
IT Information Technology

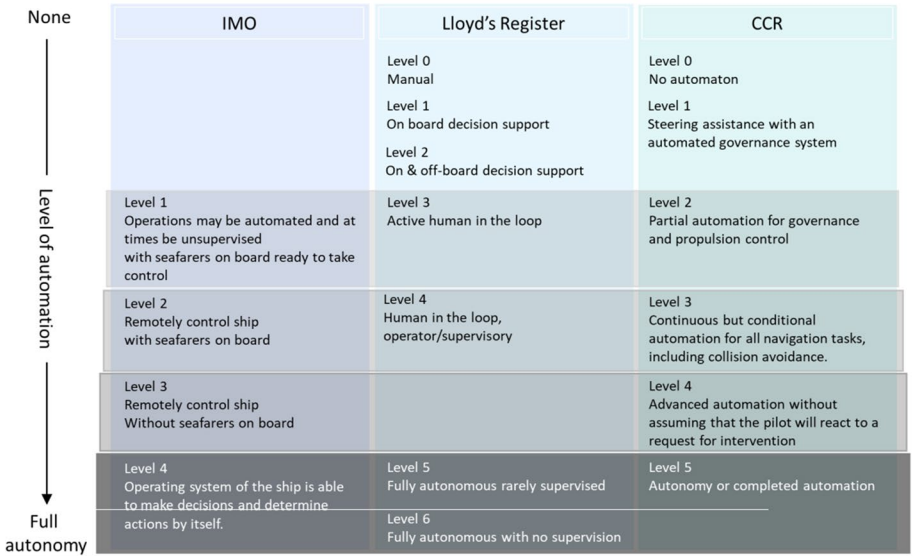
Introduction

The growing interest in maritime autonomous surface ships (MASS) reflects a broader technological and institutional shift within the maritime industry. Digitalization, artificial intelligence, and remote-control technologies are increasingly seen as tools to enhance safety, efficiency, and sustainability in maritime transport. Maritime autonomous surface ships (MASS), as defined by the International Maritime Organization (IMO) (2021), refer to commercial vessels that operate with minimal or no human intervention. Before exploring their potential implications, it is essential to clarify what is meant by "automation" in this context, since definitions vary between regulatory bodies and modes of navigation.

International Maritime Organization (*op. cit.*) defines 4 levels of autonomy and bases its classification primarily on whether control is exercised on board the vessel or remotely from shore (cf. Table 1). In the same period, Central Commission for the Navigation of the Rhine proposed the first international classification of levels of automation in inland navigation. CCNR (2018) considers 5 levels for inland navigation which it distinguishes primarily based on the level of delegation in decision-making regarding the navigation of the vessel, between the human being and the system, on the one hand, and the level of control of these decisions and their effects, on the other. Until level 2, the pilot performs some or all of the dynamic navigation tasks. From level 3, when activated, the system performs all dynamic navigation tasks. Previously, Lloyd's Register (2017) had proposed a six-level classification of autonomy in maritime transport, based on the degree to which control of the vessel is shared or delegated regardless of whether a pilot is on board.

Whatever their development and even though the level of automation may vary depending on the phase of its journey (Lynch et al. 2022), these technologies signal a potential transformation in the maritime industry, with promises to address critical challenges such as labour shortages, operational inefficiencies, and safety risks. However, these innovations also raise important concerns about their impact on the maritime workforce. While officers face the risk of displacement due to automation, ratings are expected to continue performing labour-intensive

Table 1 International Maritime Organization, Lloyd's Register and Central Commission for the Navigation of the Rhine (CCR) taxonomies of autonomy levels in water transport



tasks, such as securing cargo or maintaining the ship. This distinction has already been discussed in earlier studies on maritime automation (e.g. Schröder-Hinrichs et al. 2019), which highlight the uneven impact of automation across occupational categories. In addition, entering and exiting seaport require the supervision of a pilot who is usually on board the vessel but, in rare cases, may operate remotely as it has been recently tested in Denmark (Chambers 2025). Although container handling operations can be automated—as has been the case at the ECT Delta terminal in Rotterdam since 1993 or more recently, at Tuas Mega Port in Singapore—docking or undocking and interactions with tugboats, lashing and unlashing containers during vessel-terminal operations most often require human workers.

This paper focuses on the development of remotely controlled ships through the lens of workers and spatial considerations. It examines operational challenges posed by automation, the suitability of autonomous vessels for different shipping routes, and the motivations driving their adoption. Focusing on key research gaps, including rethinking the design of ports and redefining the role of human labour in vessel operations (Negenborn et al. 2023), this study aims to provide a comprehensive understanding of how autonomous vessels are reshaping the maritime workforce both in terms of work on board and onshore. Particular attention is given to the implications of these changes during port operations, where the interplay between automation and human labour is most visible. In parallel with regulatory frameworks, critical questions arise concerning the implications for seafarers. While autonomous systems may eliminate the need for onboard officers in certain cases, ratings are expected to continue performing labour-intensive and hard-to-automate tasks. Evidence from other sectors, such as mining, suggests that lower- and mid-skilled positions may decline as automation replaces manual work (Storey 2023). At the same time, research in the Swedish mining sector shows that workers expect a significant transformation of their roles, with a shift from physically demanding on-site tasks to more remote, office-based activities centred on monitoring automated systems and intervening when required (Tariq et al. 2025). Furthermore, these situations would outsource control of the ship, which would also have consequences for the work (Wahlström et al. 2015). These include docking, lashing, and maintenance, often carried out under unpredictable and high-stress conditions. Moreover, traditional maritime expertise remains essential in the autonomous shipping workforce, as emphasized by Belabyad et al. (2025a, b), highlighting the need for an integrated approach where conventional skills coexist with new competencies. In addition to the development of maritime autonomous surface ships (MASS), this study examines the emergence of remote pilotage, a distinct operational model where vessel navigation and control are conducted from shore-based centers. Unlike fully autonomous MASS, remote pilotage retains human oversight but introduces new challenges in terms of infrastructure, regulatory frameworks, and the division of responsibilities between on-board and on-shore personnel. This distinction is critical for understanding the broader implications of automation on maritime labor and traffic management.

This study builds upon earlier literature on the "ironies of automation" (Bainbridge 1983; Baxter et al. 2012) and addresses emerging gaps by focusing on seafarers' perspectives. By acknowledging the variability of operational contexts and the stabilizing role of regulation, it contributes to a more nuanced understanding of how automation interacts with human expertise and institutional frameworks. Our dual focus on spatial conditions and human roles complements existing technical and economic feasibility studies.

Literature review

The literature often presents automation as a desirable development in production systems. The transportation system is no exception. Automation would help to reduce ship operating costs and enhance navigation safety (Kurt and Aymelek 2022) by minimizing the occurrence of human error (Cheng and Ouyang 2021). At the same time, automation would improve performance through greater flexibility and enhanced service continuity.

However, the existing body of literature on maritime autonomous surface ships (MASS) involves diverse disciplines including engineering, ergonomics, transport geography, and maritime economics. In order to structure this diversity and avoid an impression of arbitrariness in the selection of references, this review focuses on the

consequences of MASS development on human work and its environment. It adopts an analytical framework focusing on (1) work and human factors, (2) spatial and operational conditions, and (3) economic feasibility. These three perspectives were selected because they correspond to the main dimensions through which uncertainty is produced and managed in maritime transport systems. Unlike highly controlled and closed environments—such as an automated metro line, where infrastructure, flows, and user behaviour can be tightly regulated (Foot 2005)—maritime transport operates in an open environment characterized by weather variability leading to varying and changing navigation conditions, variety of practices (work-related *vs* recreational, various professional activities involving different types of vessels in different context), heterogeneous traffic, and limited controllability under conditions of significant isolation. In the event of an accident, the crew members can only rely on themselves. This openness makes the effects of automation more difficult to predict and justifies a multidimensional reading of the literature. Within each section, references were selected based on their explicit contribution to one of these three dimensions and their relevance to recent debates on MASS.

Human factors and role displacement

As highlighted in previous studies (Bainbridge 1983), automation can lead to a paradox where the operator's role becomes more difficult precisely because he is expected to intervene only in emergency or unforeseen scenarios in the takeover situations. Autonomous systems can induce confusion in terms of modes and degree of automation if human actors are unaware of or unable to determine the level of which the system is operating (Lee et al. 2015). This is especially critical in unexpected or emergency situations. Hancke (2020) added a risk of complacency with regard to the automated system. According to the author, accidents with the Fly-by-Wire technology (FBW) developed in the aviation transport are due to the belief in and the reliance on automated systems is highly relevant on aircraft and in critical processes. Some accidents may be partly due to this overconfidence.

In maritime settings, the emergency issue is compounded by the physical distance introduced by Remote Operation Centres (ROCs), Remote Control Centres (RCCs) or Shore Control Centers (SCCs). Baxter et al. (2012) suggest that this leads to a decline in perceptive, diagnostic, and motor skills among shore-based personnel. In the unforeseen scenarios, the crew on board might overlook some warning signs, and remote pilots might ignore them. Indeed, the progressive deskilling of maritime labour threatens the industry's capacity to respond to unexpected situations, especially if future seafarers never develop the necessary experiential knowledge. Bachari-Lafteh and Harati-Mokhtari (2021) emphasize the need for a pair of expert operators involved in the operations of the autonomous ships at RCC. One will be the navigation operator in charge of management and command and monitoring in all aspects of navigation based on "ECDIS" knowledge, and principles of electronic navigation. The second will be the engine operator, responsible for monitoring, controlling and troubleshooting the ship's engine. The information quality at the SCC will be a key issue in overall performance of MASS for evaluating and controlling systems and problem solving in complex navigation situations. Situational awareness requires more than electronic data collection. Indeed, not all data is measurable. Feeling the ship's pitching movements, listening to and feeling the slamming as the bow plunges back into the wave and studying the wave patterns, contribute to the ship sense[1] (Prison et al. 2013).

This limitation is particularly important at sea, given the dynamic and open nature of the environment. Conditions change independently of human actions, and the crew members responsible for navigating the vessels must constantly adapt their decisions to these changing conditions. Human experience plays a greater role than in closed and highly instrumented systems such as metro networks. In such systems, as shown by Foot (2005) for the line 14 in Paris, automation is designed precisely to reduce uncertainty by controlling both infrastructure and users, whereas maritime automation must cope with irreducible environmental variability. Automation thus

[1] This aligns with the views of load, hand, dump (LHD) activity workers in the Swedish mining sector, who anticipate that automation may actually increase maintenance needs, as the *"the production worker loses the feel for the machine and the rock"* (Tariq et al. 2025, p. 575).

requires a reconfiguration of the socio-technical system in which uncertainty is reduced through physical barriers, controlled access, and the redefinition of interactions between engineers, operators, and users. By following how engineers and drivers conceptualize both automation and passengers, the example of the automatic metro highlights how closed systems rely on the stabilization of both technical environments and human behaviour. This difference reinforces the importance of human-centred analyses in the case of MASS.

Concerning the engine room, high cooling water temperature for engines may not necessarily be due to abnormal changes within the system but to changes in seawater temperature. Therefore, the integration of multiple pieces of information and their relevance to one's goal contributes to situational awareness, which enables decision-making to be anticipated and adjusted (Endsley 2016). If presenting a few critical information for comprehension to mitigate interface complexity and enable efficient pattern matching, caution is necessary for selecting the means of access to information less important, as it can induce attentional tunnelling and additional cognitive load, which can alternate the situation awareness and fault-detecting performance when compounded by high cognitive load from primary monitoring tasks. Support system interface (extent and timing of assistance) should be adapted to the varying operational demands (Han et al. 2024). But will that be enough? While decision-making exercises on simulator improve the ability of trainee officers on watch to analyse complex situations, they have no significant impact on the manoeuvres performed. And there is an obvious difference between trainees who experienced complex avoidance situations during their on-the-job training period and those who did not (Chauvin et al. 2009). What will happen with the adoption of MASS?

And more broadly, Tam et al. (2021) explored the psychological costs of remote operations, noting that the cognitive load on ROC operators increases during complex navigational or emergency scenarios. They further propose a hypothetical sequence of stress phases specific to operators of both manned and unmanned vessels, emphasising that prolonged exposure to multiple unconnected high-intensity tasks without opportunities for recovery may significantly heighten risks for personnel. Stress and fatigue, compounded by limited situational awareness, can negate the safety benefits touted by proponents of MASS.

Beyond the ship's steering operations, manual operations such as mooring, cargo handling, and engine troubleshooting remain difficult to automate (Kooij et al. 2018). Ratings perform many of these tasks, and automation may exacerbate divisions between officers and ratings (Lynch et al. 2022). Flécher (2014) added a crucial sociological dimension to this debate by examining how safety management systems—particularly the International Safety Management (ISM) Code—have already begun displacing the traditional autonomy of seafarers. Based on ethnographic research aboard merchant vessels, she demonstrates how formalized written procedures (e.g., checklists, risk assessments, and near-miss reports) reconfigure the role of officers from skilled practitioners into administrative intermediaries accountable to shore-based entities. These documents are not merely operational tools but instruments of control and traceability that reshape power dynamics between ship and shore. In emergency contexts, where quick and autonomous decision-making is critical, these bureaucratic mechanisms often prove inadequate, yet seafarers remain legally and morally accountable for outcomes. This shifting burden of responsibility—without corresponding operational authority—prefigures the challenges likely to intensify with the development of MASS. As human operators are relegated to roles of supervision and documentation, their ability to exercise practical seamanship may diminish, while their exposure to post-incident blame increases. Flécher's work thus underscores the risk that automation, digital oversight and remote control could further marginalize human expertise, eroding the tacit knowledge and situational adaptability that are essential to maritime safety.

Spatial constraints and operational variability

The feasibility of MASS deployment is highly context-dependent. Ahmed et al. (2023) and Munim et al. (2025) suggest that inland waterways and short-sea routes in Europe are well-suited to lower degrees of autonomy (D1 or D2), as they involve predictable conditions and short voyages. In contrast, deep-sea routes encounter numerous barriers, such as weather variability, regulatory fragmentation, and navigation through congested chokepoints.

Ports are critical nodes where automation interacts most visibly with human labour. As Negenborn et al. (2023) argue, the current infrastructure is rarely compatible with fully autonomous vessels. More generally, significant challenges remain regarding the reliability and security of communication and connectivity between ships, port systems, and control centres, particularly under variable environmental conditions. Upgrading ports involves high costs and complex engineering, especially for retrofitting docking and berthing systems.

Recent empirical research supports a differentiated geography of MASS deployment. For instance, Munim et al. (2025) demonstrate that Degree 2 autonomous feeder vessels are commercially viable in intra-European trades, particularly in the Baltic and Mediterranean Seas. These short-sea corridors, characterized by regular port calls and relatively calm sea conditions, offer a conducive environment for early adoption. The presence of standardized container handling systems and moderately advanced port infrastructure enhances feasibility. Similarly, Parvasi et al. (2025) identify the inland waterway corridor between Rotterdam and Ghent as a promising testbed for electric autonomous RoRo ships. Their case study finds that, despite higher initial capital expenditures, such vessels achieve significant operational cost savings and reduce greenhouse gas emissions by over 60%. These findings align with EU policy goals aimed at shifting freight from road to waterways, reinforcing the strategic importance of spatial and modal fit in determining the viability of MASS.

Conversely, long-haul intercontinental routes—while economically attractive due to scale economies—pose higher levels of operational uncertainty. Although large container ports along the Far East–Europe corridor have the infrastructure to accommodate technologically advanced vessels, challenges such as adverse weather, geopolitical instability, and multi-jurisdictional regulation persist. Even so, Munim et al. (2025) argue that these routes may eventually support large-scale autonomous vessels due to their high degree of standardization and volume concentration. In more remote and environmentally sensitive areas, such as Arctic shipping lanes, different operational constraints apply. Munim et al. (2022) argue that, given the extreme hazards of the Arctic environment, semi-autonomous vessels operated from shore-based control centres currently represent the most viable option. This approach significantly reduces the risk to human life by minimizing the need for onboard crew. Malmquist and Munim (2024) surveyed 37 industry and academic experts to assess the expected timeline for the adoption of maritime autonomous surface ships (MASS) across different shipping segments. Their findings indicate that regional passenger ferries are seen as the most likely early adopters within five years, followed by offshore wind farm and ro-ro shipping, while container shipping is expected to gain prominence in the 5–10 years horizon. These spatial variations suggest that the scope of MASS will be uneven and segmented, shaped by a combination of commercial conditions of routes, infrastructural readiness, and regulatory alignment.

Further geographical differences emerge when considering the operational conditions of heavily navigated waters. Rahman (2022), drawing on an analysis of 311 accident cases from IMO's GSIS records in South East Asia, shows that the prevalence of small fishing vessels and the frequent presence of drifting nets create detection challenges that current sensor technologies may not reliably address. These navigational hazards complicate the deployment of higher degrees of autonomy, as collision-avoidance systems struggle in environments characterized by dense, irregular, and highly mobile traffic. From the perspective of seafarers surveyed in the study, such conditions make fully unmanned vessels less suitable for many regional routes, suggesting that South-East Asia may require hybrid or lower-autonomy solutions tailored to its specific maritime landscape.

Economic motivations and feasibility

The adoption of maritime autonomous surface ships (MASS) is often motivated by the potential to reduce crew-related expenses, as labour costs for maintaining qualified officers onboard constitute a significant part of maritime operational expenditures. Reducing these costs is seen as a direct path to operational savings. This cost-saving potential is central to the feasibility of autonomous vessels, which involves balancing reduced crew costs against potentially higher construction costs for newbuilds designed for autonomy (Gu et al. 2021).

However, this optimistic view is tempered by several challenges. Savings from reduced onboard crew may be offset by increased expenses such as higher manning costs at shore-based Remote Operation Centres (ROCs),

elevated port fees, and substantial capital investments in MASS technologies, including cybersecurity and advanced AI systems (Hogg and Ghosh 2016). Furthermore, capital costs for MASS remain uncertain and could be 10–30% higher than conventional vessels of similar size, with added considerations like loan repayments, depreciation, registration fees, and taxes complicating the economic assessment (Munim et al. 2025). The lack of robust empirical data currently limits definitive conclusions on the cost-effectiveness of MASS adoption. Recent work by Alamoush and Olçer (2025) further refines this discussion by stressing the crucial role of maintenance operations onboard vessels. They argue that a significant proportion of these tasks would be difficult to automate within the framework of MASS deployment, which raises questions about the practical extent of crew reduction. Moreover, according to their literature review, many systems on MASS would need to be duplicated to ensure safety and reliability, a requirement that could lead to considerably higher investment and operational costs. While this perspective provides an important corrective to overly optimistic cost assumptions, it does not fully account for the heterogeneity of vessel types, operational routes, and navigational contexts. Such differences may be critical to assessing the real costs of MASS: for example, offshore supply vessels operating in controlled environments may be amenable to autonomous operations with lower costs than vessels engaged in long-distance or coastal navigation.

Case study

This study focuses on the professional perceptions of maritime autonomous surface ships (MASS) within the French maritime transport sector. It mainly concerns actors involved in maritime operations, training, and port management, and analytically focuses on how automation is perceived across different segments of the maritime system (onboard operations, port interfaces, and strategic management). In that sense it adopts a similar perspective of other studies focusing on workers' perceptions of automation, such as on the mining sector (Tariq et al. 2025), This approach allows us to capture the diversity of perspectives while maintaining coherence with three analytical dimensions studied in the literature review (work, space and economy).

Data collection and analysis

Our analysis combines a qualitative literature review with five semi-structured interviews conducted between January and June 2025. Interviewees were selected based on their expertise and direct experience with ship operations, port systems, or maritime strategy. This sampling strategy aimed at maximal variation across the three analytical dimensions identified in the literature review (work, space and economy), covering onboard operations (engine and deck departments), pilotage, offshore services, and port management. They included:

- A chief engineer turned educator at the École Nationale Supérieure Maritime (ENSM) (R1).
- A consultant, former operations manager at a ferry company, with previous experience as ship officer (R2).
- A maritime consultant, former operational reservist at CROSS, car ferry captain for twenty years, then harbor pilot at a major French port for the same period (R3) (R3).
- A naval operations coordinator; former platform supply vessel (PSV) pilot in an offshore maritime services company (R4).
- A head of strategy and development at a French RoRo port, formerly a study and then an operation manager at a major shipping company (R5).

Interviews lasted between 30 and 120 min, were conducted in person in French and were documented through detailed notes. Interviews focused on professional trajectories, experience with vessel automation, views on crew reduction, and expectations regarding MASS integration in different vessel types and contexts. Thematic coding was applied to extract patterns around operational complexity, labor substitution, and spatial appropriateness.

The interview material was analyzed using thematic analysis informed by Braun and Clarke (Braun and Clarke 2006; Clarke and Braun 2017). The analysis followed the six phases proposed by these authors: familiarization with the data, generation of initial codes, searching for themes, reviewing themes, defining and naming themes, and producing the report. Coding was performed manually by one author and cross-checked by the other author. The coding combined a deductive component—the three dimensions derived from the literature review (work and human factors, spatial and operational conditions, economic feasibility) with inductive, data-driven codes emerging from the interviews (for example, "maintenance as a barrier to crewless operation", "quayside versus onboard operation", and "career reorientation versus shore-based roles"). Codes were then grouped into three overarching themes, which structure the presentation of the results in "Operational challenges and onboard experience"–"Expected economic advantages": (1) operational challenges and onboard experience, (2) port infrastructure and automation limits, and (3) the contested economics of remote operation.

Given the small number of respondents, the study is exploratory in nature: it does not seek statistical generalization but analytical depth and the transferability of insights, in line with recent qualitative research published in this journal (e.g., Mahmud et al. 2025). Direct quotations and paraphrased interview material are used below as illustrative evidence of each theme rather than as stand-alone findings.

Operational challenges and onboard experience

Automation in maritime transport raises complex questions that go well beyond technical capability, particularly when it comes to the idea of crewless or remotely operated ships. A first theme cutting across all interviews concerns the gap between what can be technically automated and what maritime operations require in practice. Respondents consistently distinguished between routinizable navigation tasks, which they regarded as automatable, and situated judgement in degraded, congested, or unpredictable conditions, which they did not.

R1 (former chief engineer), stressed that the term "autonomous ship" was somewhat misleading—ships are already autonomous in the sense that they operate independently from land. The real issue, he argued, lay in the prospect of operating without any crew, or under remote control. While certain navigation tasks such as collision avoidance or radar monitoring could be automated, he pointed out that in degraded or congested environments—such as storms, the Strait of Gibraltar, the English Channel, Suez Canal or the Malacca Strait—these systems were not sufficient. Human judgement and physical presence remained crucial.

R4 (PSV pilot) highlighted practical limitations, noting that GPS positioning errors sometimes occur and require corrections by the pilot on board.

A closely related sub-theme concerns maintenance as an irreducible human task. R1 also stressed on the importance of distinguishing between commercial operations (like ballast and tank management in particular with the change in tank pressure depending on the time of the day and the outside temperature) and technical operations, particularly maintenance. Maintenance at sea, he explained, was highly variable and often unpredictable. While some minor failures might be handled by automated systems, many interventions required the expertise and improvisation of trained personnel—skills difficult to replicate remotely. This finding aligns with evidence from other sectors such as mining (Tariq et al. 2025). In addition, R3 (maritime pilot) alerted on the fact that major maintenance operations result in a loss of maneuverability. For this reason, they must be carried out on the high seas in areas with low traffic in order to avoid any risk of collision, but it is also in these areas that the adoption of automation could be most feasible in the case of long-distance navigation, as R3 (maritime pilot) suggested it. Drawing on his experience and knowledge of industry practices, he noted that some companies, like CMA CGM (pure carrier), used specialized maintenance teams that moved between vessels, while others, such as Esso-Exxon (shipper), relied more heavily on permanent onboard crews. In his view, fully engineerless ships remained unrealistic due to the complexity of onboard systems and the frequency of small but critical technical issues.

In terms of career trajectories, R1 acknowledged that many seafarers eventually sought a transition to shore-based roles, but he remained skeptical about the appeal of remote ship operation as a meaningful alternative. Maritime training continued to attract students—both men and women—who aspired to sail rather than manage

ships from land. He also challenged the assumption that remote navigation might be more appealing to women for reasons of work-life balance, noting that female students at ENSM showed the same desire to go to sea as their male peers.[2] Meanwhile, the rise in demand for officers—particularly in the wake of Brexit—had led the French government to increase maritime training capacity, signaling the continued relevance of seafaring professions.

Looking ahead, R1 considered that partial automation might be feasible on certain routes, such as transatlantic tanker voyages, where crews could be deployed by helicopter for port departures and arrivals, with the open-sea segment managed remotely. However, such scenarios would require significant investments in system redundancy, enhanced diagnostics, and new legal frameworks. More complex ships—container vessels, gas carriers—remained far less suited to full automation due to their operational demands. In the case of passenger ships, the presence of large numbers of people on board raises critical safety and evacuation concerns that make full automation particularly challenging, although recent experimental projects have begun to explore unmanned passenger operations, including trials on a small passenger vessel (Kokubun 2022). Even bulk carriers, which appeared more compatible with remote control, still presented risks such as cargo liquefaction or load shifting that required human oversight. In his view, the future of maritime automation would depend as much on legal, economic, and human factors as on technological progress.

Taken together, these accounts locate the boundary of automation not in navigation technology itself but in situations requiring improvisation, embodied experience, and physical presence. In this respect, they provide empirical support, from the perspective of practitioners, for the "ironies of automation" identified by Bainbridge (1983) and for the importance of "ship sense" (Prison et al. 2013) discussed in "Human factors and role displacement" section.

Port infrastructure and automation limits

A second theme concerns the spatial asymmetry of automation: respondents described automation as advancing on the quayside while remaining limited onboard, and as supporting rather than replacing dock labour. R5 (head of strategy, RoRo port) highlighted that automation is progressing on the quayside but remains limited onboard. In a port like Calais, where operations are highly standardised and vessel calls are frequent and tightly scheduled, certain tasks—such as bridge positioning, gangway control, and twistlock handling—are increasingly being automated to improve safety and reduce turnaround times. However, these advances often support rather than replace human workers, particularly where irregularities or disruptions occur.

Labour organisation in RoRo terminals remains largely dependent on predictable timetables and long-standing coordination between port operators, shipping lines, and dockworkers. The automated systems introduced in recent years—such as automatic mooring hooks and guided bridge levelling—have streamlined some operations, but dockworkers continue to play a central role in vehicle loading, securing, and supervision. Automation in this context is seen more as an aid to efficiency and safety than a direct substitute for human labour.

The port's transition towards electrification—spurred by the introduction of fully electric ferries—adds another dimension to the automation debate. Shore-side power supply (OPS) and standardised charging operations could, in time, further automate vessel-port interactions. However, these developments depend on heavy infrastructure investment and close coordination with electricity providers, especially as power demand increases significantly. The current grid capacity is already nearing its limit, and expanding it requires substantial upgrades to substations and cabling systems.

[2] As kindly suggested by one of the reviewers, this perspective should be nuanced. While initial motivations to go to sea appear similar across genders, evidence suggests that career trajectories are often shorter for navigating officers, with many—both men and women—seeking shore-based positions after only a few years at sea. In addition, structural constraints, including work-life balance challenges and limited access to senior roles, may further encourage women to transition earlier to land-based maritime employment. For detailed analyses, see Kitada (2020) on barriers to the employment of women seafarers and Iskra (2007) on discrimination against women on board ships.

Despite these technological shifts, R5 stressed that automation is unlikely to eliminate the need for human labour in RoRo terminals any time soon. Instead, the emphasis is on re-skilling and adapting existing roles to support a more digitally integrated and energy-efficient port environment. For the moment, human oversight remains essential in managing exceptions, ensuring safety, and maintaining the pace and reliability expected in a high-frequency port like Calais. However, this reflects a specific operational context: unlike major Asian ports operating continuously around the clock, traffic in Calais is more cyclical, which may limit the scope and pace of automation. This theme is consistent with the literature identifying ports as the nodes where automation and human labour interact most visibly (Negenborn et al. 2023), while adding that the pace and scope of automation are conditioned by local traffic patterns, labour organisation, and infrastructure capacity.

Expected economic advantages

A third theme concerns the contested economics of remote operation. While cost reduction is the dominant rationale for MASS in the literature ("Economic motivations and feasibility" section), respondents questioned whether the expected savings would materialize once shore-based staffing, equipment, and maintenance are taken into account. From an economic standpoint, automation is often justified by reducing labour costs. R2 (former chief officer) estimated crewing costs at around $130,000 per officer per year. However, R2 questioned whether remote operation can truly generate savings. A single shore-based officer might monitor multiple ships, but this depends heavily on situational complexity and the quality of onboard technology. Crucially, remote control would require expensive equipment, software, and maintenance, potentially nullifying the expected savings. He emphasises that current automation projects often underestimate the real labour burden on board, particularly in terms of maintenance, safety, and ship upkeep.

The daily work of seafarers—rust removal, painting, greasing, refrigerated container checks, and catering—remains essential. Even if some functions like navigation or mooring can be automated, the volume of manual tasks aboard cannot yet be outsourced or mechanised efficiently. The interviewee argues that removing navigational crew could break vital feedback loops between human observation and ship functioning. Rather than displacing seafarers entirely, automation might lead to a redistribution of tasks on board or a shift toward land-based maritime jobs, appealing to younger generations less inclined toward long-term seafaring careers.

While R2 and R4 (PSV pilot) acknowledged niche opportunities for full autonomy—such as offshore supply barges that remain close to platforms and are relatively static—most vessels (e.g., tankers, passenger ships, cable layers) are not viable candidates for full automation. Despite growing attention to automation, recruitment challenges and a general trend toward career diversification away from sea-going roles suggest that shore-based positions in remote operations could gain popularity. Still, R2 cautioned against over-investment in speculative technologies driven by funding trends rather than operational need or feasibility.

These findings should be interpreted cautiously given the limited number of respondents, but they highlight recurring concerns regarding the three analytical dimensions studied in the literature review (work, space and economy).

Discussion

The findings reinforce and deepen existing critiques of automation's uneven impact on maritime labour. MASS may not replace labour in a uniform way; instead, they reshape the geography of labour and introduce new forms of spatial and social inequality. The interviews confirm that MASS is most feasible in controlled settings (offshore, inland, or short-sea routes) but remains limited in complex environments such as ports or transoceanic shipping in geographical areas with heavy traffic and the presence of different types of activity.

This supports the call for differentiated regulatory and training frameworks, as well as a potential reorganization of maritime work tasks, akin to the aviation sector. Rather than a one-size-fits-all model, we advocate for

context-specific strategies. This includes tailored investments in port infrastructure, ergonomic design of remote workstations, and hybrid training for officers and ratings that combines IT proficiency with operational resilience.

In this context, remote pilotage emerges as a distinct operational model, separate from fully autonomous MASS. Offshore service vessels, such as those supplying oil platforms, are among the most promising candidates for autonomous or unmanned operation, according to the interviewee. Their operating conditions—limited geographic range, predictable routes, and long idle periods—make them easier to supervise remotely compared to ocean-going cargo ships. In such contexts, remote pilotage could allow a shore-based operator to monitor multiple vessels, but this requires addressing the unique challenges of ship inertia, continuous monitoring, and team coordination—issues not present in aviation.

The oil and gas sector also has the financial and technical capacity to support this shift, as it already uses advanced systems like dynamic positioning and remote monitoring on offshore installations and vessels. Investing in unmanned support ships would build on existing infrastructure and could help reduce human exposure to risky environments. R4 stressed that such vessels are a realistic near-term application for full autonomy, especially where functions are repetitive and safety risks can be managed through remote intervention or occasional on-site maintenance. In the absence of the entire crew on board, preventive maintenance should also be prioritized over corrective maintenance that may be necessary during transport. However, the development of preventive maintenance incurs costs due to the need for more frequent shutdowns and de facto reduction in transport capacity availability.

"Collaborative autonomy" where "automation plays the role of an assistant to human operators, performing low-level data-intensive tasks, but leaving decisions that would be considered critical in human hands (the augmenting path)" though an adaptative strategies (Pöhler et al. 2025). Training and skills retention must be maintained to ensure that humans agents remain competent, particularly with regard to their ability to anticipate (Chauvin and Lardjane 2008), even as automation increases.

To maintain reliability and safety in maritime transport, large-scale introduction of autonomous ships will require "traffic-control centres". This aligns with literature, which emphasize MASS potential on long-haul intercontinental routes, but also acknowledges significant challenges such as increased traffic management complexity in high-density shipping lanes and potential communication failures in open waters (Munim et al. 2025). These challenges demand robust traffic control infrastructure and remote operations oversight.

Coastal and short-sea routes present additional complications. Congested waterways host multiple vessel types conducting different activities, creating detection issues that existing maritime systems struggle to manage (Rahman 2022). Yet regulatory frameworks for such controlled environments remain underdeveloped. Should traffic control be solely the responsibility of ship operators, or does it require higher-level oversight?

The International Civil Aviation Organization (ICAO) offers an instructive model. For over 75 years, ICAO has managed highly complex, high-speed critical operations in regulated airspace where hundreds of competing companies operate (Waldo 1947). A similar centralized governance approach may be necessary for autonomous maritime operations.

To explore the potential applicability and limitations of aviation governance to maritime autonomous operations, we conduct a comparison across five dimensions, which we consider will help understand the three dimensions that have guided our investigation (spatial, organizational, and operational): (1) regulatory structures; (2) spatial division and zoning systems; (3) traffic control objectives and protocols; (4) organizational implementation at operational levels; and (5) decision-making responsibilities in critical situations. This comparison does not emerge directly from our interview data but rather from the literature and our previous work (Dusire and Mundutéguy 2000). This is an analytical framework that we introduce to test whether aviation governance offers transferable lessons for maritime autonomous systems. This comparison serves three purposes: first, to identify features that have proven effective in aviation and may be adapted to maritime contexts; second, to highlight differences that constraint direct transfer; and third, to clarify what a maritime governance model would realistically need to look like to support MASS and remote pilotage while accounting for maritime-specific complexity.

The Table 2 shows parallels and differences. On the surface, both systems employ hierarchical regulatory structures (ICAO/FAA for aviation, IMO/national maritime authorities for shipping), spatial division into managed zones, and dedicated traffic management services. This organizational similarity might suggest that maritime governance could be modelled directly on aviation.

However, three notable differences severely constraint such direct transfer:

Institutional homogeneity Aviation operates under near-complete standardization, with commercial aircraft built by fewer than 30 manufacturers worldwide, all operate under common standards (ICAO annexes), and operate in tightly regulated airspace. Maritime transport, by contrast, involves over 1200 shipyards, vessel types ranging from 100-tonne coastal craft to 400,000-tonne bulk carriers, and a diversity of operational contexts (international high seas, territorial waters, inland waterways, port approaches) with vastly different regulatory requirements. Standardizing to the degree achieved in aviation is therefore neither technically feasible nor economically viable.

Spatial control and regulation Aviation operates in bounded, engineered airspace where infrastructure (beacons, airways, radar coverage) is installed and maintained by state authorities, enabling coherent surveillance and control. Maritime space, particularly high seas beyond the 200-nautical-mile exclusive economic zone, remain basically unregulated and ungoverned. States coordinate through international law (UNCLOS) but cannot enforce geometric separation or prescribed routes. This renders the ICAO model of continuous airspace surveillance more difficult to adapt to open-ocean MASS operations.

Attention and cognitive demands of remote operation Partially remote air traffic control works, because within the aircraft the pilot and a co-pilot maintain predictable flight path proposed by air traffic controllers, cruise at constant altitudes in controlled corridors, and experience long-duration cruise phase where workload is cyclical. Maritime remote pilotage must face continuous environmental variability (weather, currents, sea state), unpredictable vessel dynamics (cargo shifting, engine issues, structural stress), interaction with both autonomous and conventional traffic, and the need for 24/7 continuous monitoring of multiple vessels with limited opportunity for workload recovery (Chang et al. 2019). Interview data (R4, PSV pilot) suggested that even in relatively constrained contexts (offshore service vessels), the cognitive load on remote operators increases sharply during complex manoeuvres or emergencies, raising questions about the generalization of 1:N (one operator: multiple vessels) models that functions routinely in air traffic control.

While aviation provides useful organizational precedents for autonomous systems governance, maritime MASS seems to require a very different approach. Rather than a centralized authority inspired by ICAO, MASS governance will probably remain segmented and context-specific: short-sea vessels in controlled waters could operate under standardized frameworks, while ocean-going traffic in geopolitically sensitive or congested areas requires flexible, coordinative structures. Remote Operation Centres and remote pilotage, though operationally promising for specialized vessels in bounded contexts, are unlikely to replicate aviation's success for general-purpose shipping, as maritime decision-making and perceptual demands exceed current capabilities. Critically, the political reality of distributed maritime authority, fragmented across the IMO, flag states, port authorities, and coastal state, limits the kind of centralized regulatory power that ICAO has. Instead, MASS governance is more likely to evolve as a hybrid model: IMO framework-setting at the global level, delegation to regional implementations, and operational responsibility retained by flag states and remote-control centre operators. To ensure the safety of the transportation system, coordination efforts across the various levels of control will be necessary.

Conclusion

The transition toward maritime autonomous surface ships (MASS) is shaped as much by institutional, spatial, and human factors as by technological progress. Our study contributes to the literature by providing empirical evidence on the fact that automation in maritime transport is not merely a technical or cost-reduction issue, but a socio-technical transformation that reconfigures work, responsibility, and spatial organization. Interviews with

Table 2 Comparison of transport operations management in airspace and maritime space

	Airspace	Maritime space
Instances (International and National scales)	International Civil Aviation Organization (ICAO) Federal Aviation Administration (FAA) for USA, *Direction Générale de l'Aviation Civile* (DGAC) for France	International Maritime Organization (IMO) Service hydrographique et océanographique de la Marine (SHOM-for France
Space division	Airspace areas: Flight information region (FIR) [from 0 to 195 10″ feet, in France] managed by an Area Control Center (ACC) or air route traffic control center (ARTCC) Controlled Traffic Region (CTR) and Terminal Manoeuvring Area TMA [0–195] Lower Traffic Area–(LTA) [115–195], referring to a controlled space Upper Traffic Area (UTA) [195–660], Class C controlled space [>660] Uncontrolled airspace in UIR UIR [>FL 195]	Maritime zones (saltwater areas, in free and natural communication): Inland waters, between the shore and the baseline formed by the low-water mark Territorial sea (from the baselines to 12 miles) Contiguous zone [12–24 miles] Exclusive economic zone [200 miles] High seas [>200 miles]
Objective	The airspace division ensures safety by facilitating air traffic control, adapting rules according to traffic density and type, and ensuring consistent use of airspace	The division of maritime space aims to reduce conflicts between states and stakeholders by defining areas of influence and exploitation, encouraging cooperation between maritime actors for sustainable resource management, and ensuring better protection of the marine environment through the creation of marine protected areas. From an economic perspective, this division encourage investment by offering legal certainty to stakeholders, develop renewable energies and other maritime activities (maritime transport, laying of submarine pipelines and cables) and strive to reduce the impact of human activities on ecosystems
Traffic control (International and/or National scales)	Air Traffic Control	Vessel Traffic Services (VTS)[a] or Navigation Center (NAVCEN) in USA, and *Centres régionaux opérationnels de surveillance et de sauvetage* (CROSS) in France
Organisation	Air traffic control regulate traffic in the corridors of each sector in LTA and UTA Approach control regulates air traffic in the airspace of an airport terminal (CTR) to ensure the safety of aircraft climbing to their cruising altitude and those descending to land	Control centres monitor traffic in busy corridors and pollution, disseminates information for safety, organises rescue operations The entry and exit phases of a port are carried out under the control and guidance of a pilot, who boards the ship
Objective	Air traffic controllers analyse traffic and issue clearances for flights in their sector, provide flight instructions (altitude, direction) to pilots entering the airspace, and monitor their progress in the sector for which they are responsible	Agents at the navigation centres monitor maritime navigation and pollution in the sector for which they are responsible, transmit information with a view to maintaining maritime safety, and organize rescue operations
Operational management (Company level)	Airline Operations control Centre	Fleet Navigation and Support Centre
Organization	The agents monitor the development of all the flights, operated daily by the airline and its partners around the world, 365 days a year, 7 days a week, 24 h a day	Based on live analytics of the vessels' operating speeds, ocean currents, weather forecasts and high traffic areas, the agents assess, anticipate and mitigate any navigation-related risks to the operations
Objective	Optimisation of capacity, management of contingencies and reallocation of resources in the event of capacity loss	Optimization of vessel routes, speed and fuel consumption, management of unforeseen events to ensure the safety of the crews and cargo, and reallocation of ports

Sources: Airspace column derived from ICAO and FAA. Maritime column follows United Nations Convention on the Law of the Sea (UNCLOS 1982), IMO regulations on Vessel Traffic Services (IMO Resolution A.857(20)), and national maritime authorities. The comparison of institutions, objectives, and organizational implementation is the authors' synthesis, informed by literature on maritime governance (Munim et al. 2025; Ahmed et al. 2023) and remote operations (Tam et al. 2021; Bachari-Lafteh and Harati-Mokhtari 2021)

[a]VTS play a critical role in managing maritime traffic, especially in high-density areas. With the rise of MASS, VTS may need to evolve to include new functions such as remote monitoring, conflict resolution between autonomous and manned vessels, and integration with shore-based control centers

seafarers and port professionals highlight that human presence remains indispensable in degraded, congested, or unpredictable environments, particularly for navigation, maintenance, and other tasks requiring expertise and improvisation. Even partial automation is constrained by the variability of operational conditions and the need to maintain safety and continuity.

Unlike previous studies that focus on technical feasibility or economic justifications, this research highlights the nuanced, context-dependent nature of automation adoption. At the port interface, automation primarily functions as an aid to efficiency and safety rather than a replacement for human labour. Tasks such as mooring, gangway management, and vehicle loading may be partially automated, but human coordination and oversight remain central, particularly when disruptions occur. Our findings challenge the assumption that automation will uniformly reduce crew numbers, instead showing that it is more likely to redistribute responsibilities between ship and shore, while creating new roles and demands for hybrid skills.

A key takeaway from this study is the identification of operational, spatial and economic issues that determine the feasibility of automation. For example, offshore service vessels with limited geographic range and predictable routes are more amenable to remote pilotage, while complex environments like ports or heavily congested areas would be resistant to full automation. This contributes to a new framework for understanding where and how automation can be viably implemented, based on empirical evidence from French maritime stakeholders. Additionally, our analysis reveals that economic expectations of crew reduction are often overstated, as the persistent need for maintenance, diagnostics, and manual operations on board offsets potential savings from remote monitoring.

This study also advances the discussion on the organizational and regulatory implications of MASS. The comparison with aviation (e.g., ICAO) provides useful organizational precedents but reveals that maritime autonomous governance cannot simply replicate the aviation model. Instead, maritime MASS regulation will likely remain segmented and context-dependent, with full autonomy and remote pilotage feasible only in relatively controlled operational contexts (short-sea routes, offshore service vessels, predictable corridors), while complex coastal and congested environments will require hybrid governance incorporating human oversight, adaptive regulatory frameworks, and continued emphasis on maritime expertise alongside technical capability.

Nevertheless, this study presents several limitations. The empirical analysis is based on a small number of respondents, which limits the representativeness of the findings and calls for caution in generalizing the results. In addition, the research mainly focuses on French maritime stakeholders and selected operational contexts, while the development and acceptance of automation may differ significantly across countries, regulatory environments, and maritime sectors. Future research should therefore broaden the geographical scope of the analysis and include a wider range of vessel types, such as container ships, tankers, ferries, or fishing vessels, in order to provide a more comprehensive understanding of the operational, social, and economic implications of MASS.

Our research provides recommendations for industry and policymakers. Automation should not be approached merely as a cost-reduction mechanism but as a means to retain critical human expertise, reduce operational stress, and account for route-specific and geographical particularities. Future research should expand empirical evidence through larger interviews and quantitative surveys of active and prospective seafarers to better understand job expectations, skill gaps, and acceptance of emerging technologies. While hazard and safety issues have been addressed elsewhere (Sumon et al. 2025), this study uniquely emphasizes the social, spatial, and organizational transformations induced by MASS, highlighting the need for hybrid systems that integrate human adaptability with technological capabilities.

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Declarations

Ethics approval and consent to participate This study was conducted in accordance with institutional and national ethical guidelines. Participation in the semi-structured interviews and questionnaires was voluntary, and informed consent was obtained from all participants prior to data collection.

Consent for publication Not applicable.

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